INCH-POUND

A-A-54738
11 September 1992

COMMERCIAL ITEM DESCRIPTION

PIPETTING MACHINE, SEMI-AUTOMATIC

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-P-36637B which is cancelled.

This Commercial Item Description covers two sizes of manually operated pipetting machines used in the dispensing of repetitive amounts of reagent solutions in laboratory procedures.

Salient characteristics:

General. The pipetting machines shall be of the following sizes:

Size	Description
1	0-10 milliliters (ml) range (30 ounces).
2	0-1 ml range (16 ounces).

Materials, style, and design. The pipetting machine shall consist of a bottle, cap, vent, and measuring and dispensing assembly and shall be in accordance with Figures 1 or 2.

Bottles. Bottles shall be fabricated of glass. Neck shall be threaded for easy removal and closure of the cap. Color of bottle shall be uniform for each contract.

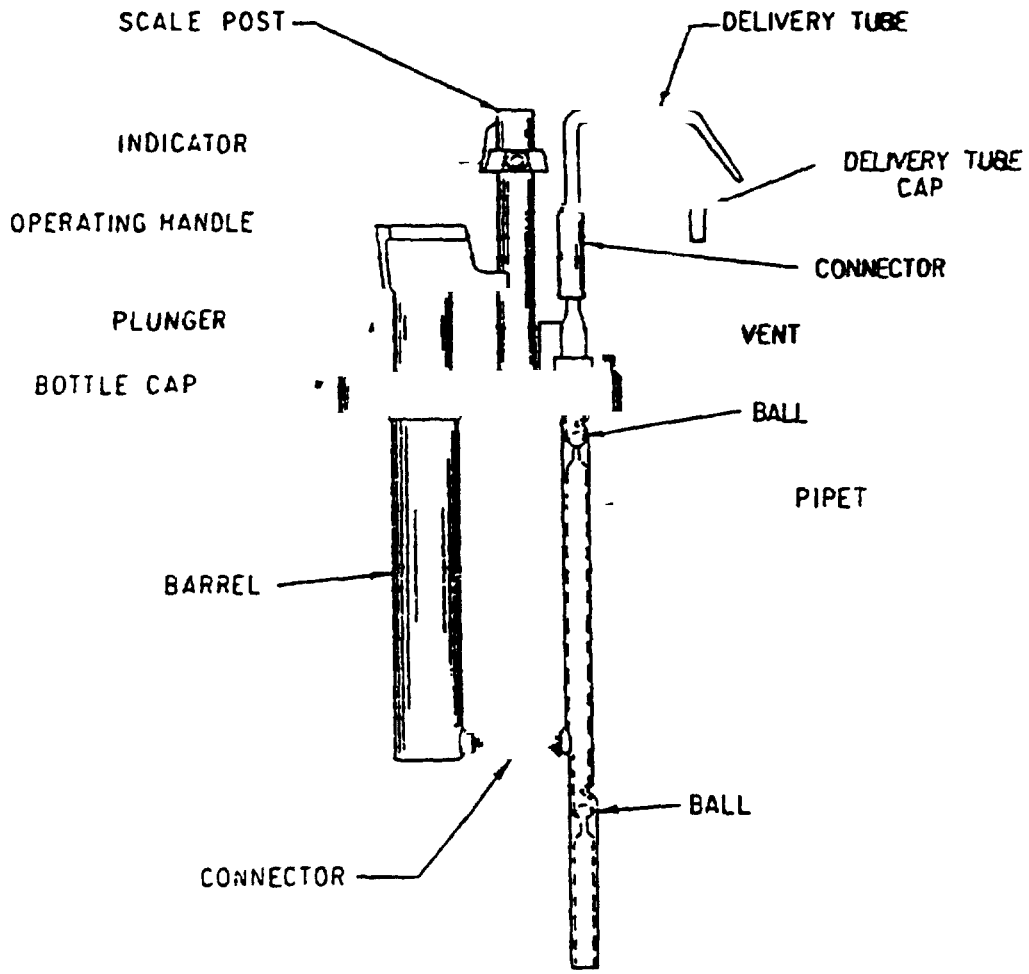
Cap. Cap shall be fabricated of plastic and shall be threaded to mate with the neck of the bottle. It shall have sufficient holes to permit passage of functional parts through the cap.

Measuring and dispensing assembly. Measuring and dispensing assembly shall consist of a plunger and barrel assembly, a ball valve assembly, a delivery tube, and a scale with an adjustable indicator.

AMSC N/A FSC 6640

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

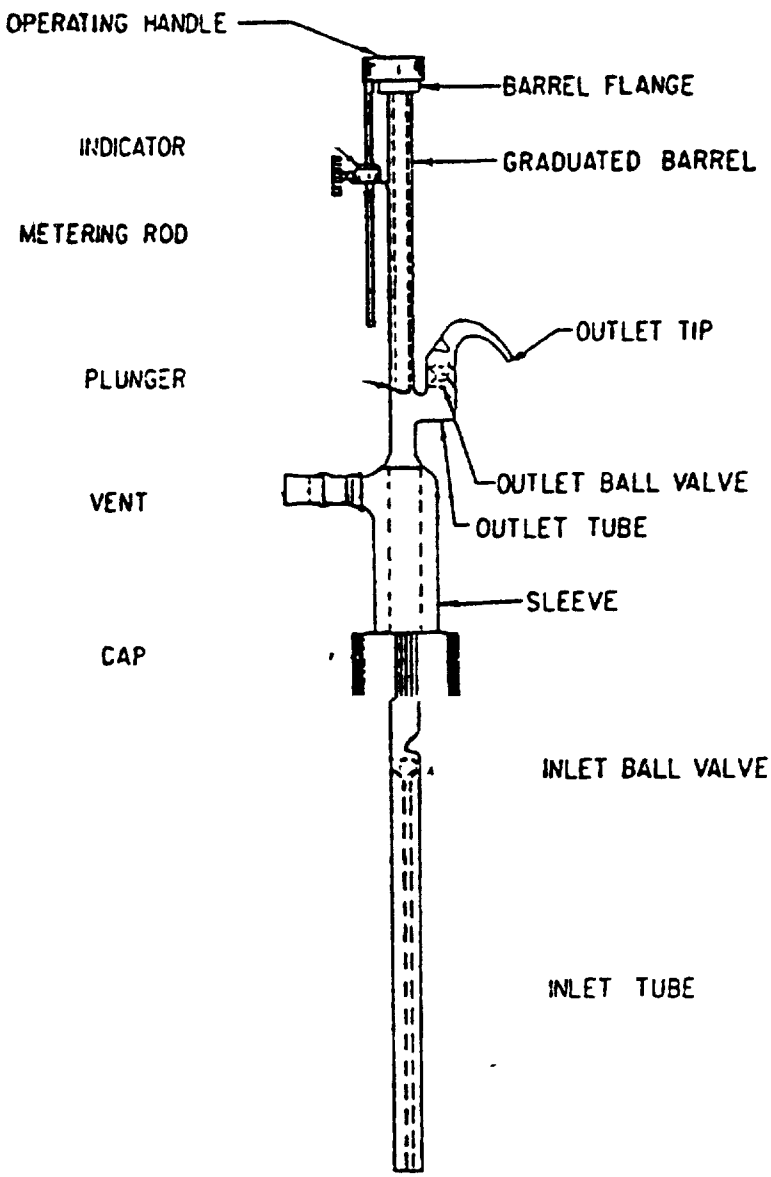
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FIGURE 1 PIPETTING MACHINE

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FIGURE 2 PIPETTING MACHINE

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Plunger and barrel assembly. Plunger and barrel assembly shall be fabricated of glass. Top of plunger shall have a plastic or aluminum operating handle. When furnished, metering rod (Figure 2) shall be corrosion-resisting steel or glass-filled plastic.

Ball valve assembly. Delivery shall be accomplished by means of the action of two ball valves.

Delivery tube. Delivery tube shall be fabricated of glass. A plastic cap which fits over the tip of the delivery tube shall be furnished. Cap shall be attached to the delivery tube by means of a flexible plastic ring and connector.

Indicator. Indicator shall be fabricated of plastic or aluminum. Means shall be provided for placing and securing the indicator at all applicable scale graduations specified in Table I.

Scale. Scale shall be marked on a separate plastic scale post (Figure 1) or on the exterior of the barrel or plunger (Figure 2).

Capacity and graduations. Bottle capacities shall be as specified in Table I. Graduation markings shall be as specified in Table I. Smallest graduation shall not exceed that shown in Table I.

TABLE I. Capacities and graduations

	Size 1	Size 2
Bottle capacity in ounces minimum	30	16
Graduated range in milliliters	0-10	0-1
Smallest graduation interval in milliliters	0.2	0.02
Numbered graduations in milliliters	0,1,2,3,4,5,6,7,8,9,10	0,.1,.2,.3,.4,.5,.6,.7,.8,.9,1.0

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Performance. The pipetting machine shall be tested for precision and accuracy.

Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts.

Marking. Graduation markings shall be permanent and legible. Each pipetting machine shall be legibly marked with the name or registered trademark of the manufacturer.

Workmanship. Workmanship shall be first class throughout. Pipetting machines shall be free from breaks, cracks, chips or deformation or any other defects which detract from its appearance or serviceability. Missing components or service data shall constitute a defect.

Unit. Each (EA). One pipetting machine, of the size specified, constitutes one unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one pipetting machine.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests. Sampling for tests (dimensional) shall be conducted in accordance with MIL-STD-105 with an AQL of 1.0 (percent defective) and an inspection level of S-1. The unit of product for test purposes shall be one pipetting machine.

Accuracy.

Place receiving vessel on balance pan and tare.

Fill reservoir with the water and ensure inlet and outlet tubing are correctly installed.

Set dispenser to desired volume.

Prime dispenser until no air bubbles are present in outlet tubing.

Open balance door and position outlet tubing over receiving vessel.

Gently raise the plunger assembly until it stops. Pause, then dispense sample by gently pushing down on plunger assembly until it bottoms out. Touch off any partial drop remaining on end of outlet tubing against inner wall of receiving vessel.

Repeat above to obtain 10 replicants, making sure the balance is tared before each dispensing.

Calculate the precision and accuracy using the procedure below.

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Calculating precision (P).

Calculate the mean sample($\bar{S}$) and the standard deviation (S.D.).

Calculate the precision using the following formula: Precision should not be more than 1%.

$$P = \frac{S.D.}{\bar{S}} \times 100\%$$

Calculating for accuracy (A).

Calculate the accuracy using the following formula:

$$A = \frac{\bar{S}\text{-Intended Volume}}{\text{Intended Volume}} \times 100\%$$

Accuracy should not be more than $\pm$1%.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered material. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

NSN	Item Identification
6640-00-176-5244	PIPETTING MACHINE, SEMI-AUTOMATIC, Manually Operated, 0-10 ml Reagent Range.
6640-00-176-5245	PIPETTING MACHINE, SEMI-AUTOMATIC, Manually Operated, 0-1 ml Reagent Range.

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MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6640-1764

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER A-A-54738	2 DOCUMENT DATE (YYMMDD) 11 SEPTEMBER 1992

3 DOCUMENT TITLE
PIPETTING MACHINE, SEMI-AUTOMATIC

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a NAME (Last, First, Middle Initial)	b ORGANIZATION	
c ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7 DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete* 198/290

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